BY ORDER OF THE COMMANDER
JOINT BASE LANGLEY-EUSTIS (ACC)

JOINT BASE LANGLEY-EUSTIS
INSTRUCTION 36-101

10 DECEMBER 2020

Personnel

PRIVATELY OWNED FIREARMS RANGE PROGRAM (FORT EUSTIS ONLY)

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available on the e-Publishing web site at www.e-publishing.af.mil for downloading or ordering.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: 733 FSD

Certified by: 733 MSG/CC
(COL Chesley D. Thigpen)

Supersedes: JBLEI 36-101, 18 March 2014

Pages: 25

This publication implements the Privately Owned Firearms (POF) Range Program at Fort Eustis, Virginia. It applies to all persons eligible to fire their own firearm on Fort Eustis. Ensure all records created as a result of processes prescribed in this publication are maintained IAW Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. Submit requests for waivers through the chain of command to the MAJCOM/CC with the concurrence of the publications Approving Official. This instruction requires the collection and maintenance of information protected by the Privacy Act of 1974. The authority to collect and maintain the records prescribed by this instruction is 10 United States Code (USC) § 8013. Each form, format, or form letter prescribed by this instruction that requires a Privacy Act Statement (PAS), in accordance with AFI 33-332, *Air Force Privacy and Civil Liberties Program*, must have the PAS incorporated thereon or the PAS will be prominently posted and be provided by the requester prior to collecting this information. Privacy Act information taken from an existing privacy act system of records, which authorizes blanket use of this information for this purpose, does not require a PAS. Collected information is "For Official Use Only". Requests to release Privacy Act Information to persons or agencies outside the Department of Defense (DoD) must be in accordance with AFI 33-332. The applicable System of Records Notices (SORN) F031 AF SF B, *Security Forces Management Information System*, and F031 AF SP O, *Documentation for Identification and Entry Authority*, apply and are available at http://dpcld.defense.gov/Privacy/SORNs/.

SUMMARY OF CHANGES

There has been no major changes to this publication. Publication should be reviewed in its entirety because references and prescribed forms have been updated.

Chapter 1

GENERAL INFORMATION

1.1. Purpose. This instruction prescribes policies and procedures for use of the Privately Owned Firearm (POF) Range on Fort Eustis, Virginia.

1.2. Scope. This publication applies to all personnel that utilize the Fort Eustis POF Range**.** The Fort Eustis POF Range is available for use to all Active Duty Military, Guard and Reserve, Family members, retirees, veterans with base access, and DoD civilians and contractors with unescorted access to the installation, who wish to fire their POF on Fort Eustis firing range.

1.3. Administrative. The 733d Force Support Division (FSD) Outdoor Recreation (ODR) is the OPR for management of this program. Persons utilizing this program must:

1.3.1. Register firearms at the Base Defense Operations Center (BDOC), Bldg. 648 Washington Blvd., using the AF Form 1314, *Firearms Registration*. The DD Form 2760, *Qualification to Possess Firearms or Ammunition*, must accompany the registration at the time of drop off. Forms can be picked up from the BDOC or downloaded at https://www.e-publishing.af.mil or http://www.dtic.mil/whs/directives/forms/dd/ddforms2500-2999.htm. For more information about the registration process, contact the BDOC at (757) 878-4555/4556. Only firearms outlined in Chapter 3 will be registered for use on the range.

1.3.2. Sign the Safety/Range Rules (Attachment 2) and Release and Hold Harmless Agreement (Attachment 3 and 4). Eligible persons utilizing the range are doing so at their own risk. These documents will be included in the Safety Brief and kept on file at the ODR Rental Office.

1.3.3. Purchase a POF Range Permit. The POF Range Permit can be purchased at ODR, 828 Kells Dr., Fort Eustis, Monday through Friday, 0900-1600. Minors and non-DoD affiliated persons cannot be issued a card.

1.4. Location and Hours of Operation.

1.4.1. The POF Range is located at 3909 Mulberry Island Road, Fort Eustis. Range #3 will be used for all authorized firearms IAW Chapter 3 of this publication.

1.4.2. The POF Range is operational alternate weekends (weather permitting). The hours of operation are Saturday and Sunday, 0900-1300. Dates and times will be posted on the 733 FSD website, www.jble-eustismwr.com. Range priority will go to scheduled military operations and to other government activities when required.

1.5. Emergencies. In case of emergency requiring Fire, Medical, and/or Security Forces response, dial 9-1-1 to contact Fort Eustis Emergency Services. Give the operator name, location, nature of injury, and how many persons are involved. Do not hang up until told to do

so by the operator. Leave all firearms and equipment in place unless it is necessary to move them to provide aid. Remain on the scene to guide first responders and provide information to Security Forces personnel. Depart only when allowed by Security Forces. If someone leaves before first responders arrive, contact the BDOC at (757) 878-4555 and give a description of the person and/or vehicle; do not attempt to stop the person(s). The 733 FSD Range Safety Officer (RSO) will ensure all appropriate reporting is completed for the 733 FSD Safety Representative and Range Control.

Chapter 2

POF RANGE OPERATIONS

2.1. Use. Use of the POF Range is a privilege. Personnel that utilize the range will conduct themselves accordingly. Failure to follow instructions contained in this publication could result in withdrawal of privileges.

2.2. Range Sign-In. The 733 FSD RSO will coordinate with Felker Army Airfield (FAAF) Flight Operations at (757) 878-5828/2584, NLT the Friday prior to weekend POF operations and request a "Notice to Airmen" (NoTAM) to be published reflecting dates and times of planned range operations. The 733 FSD RSO will be posted at the POF Range during hours of operation and will be responsible for all range operations. All patrons will sign-in with the RSO and sign-in any guest(s) that are accompanying authorized patron at the POF Range. The RSO is responsible for ensuring any and all patrons that show up to the range have signed in. When signing in, the RSO must be presented with an AF IMT 483, *Certificate of Competency*, POF Range Permit, the type of firearm(s) to be fired and the number of rounds expected to be fired on the range from each firearm.

2.3. Range Safety Officer (RSO).

2.3.1. The RSO will be identified by name tag, and a florescent orange/green vest/shirt will be worn on the POF Range. The RSO will check the range prior to opening for any safety violations and complete the opening checklist. After checking for safety violations and excess shells, the RSO will raise the range flag. The RSO will also be responsible for reporting any violations of this publication and unsafe acts conducted on the range. On the occurrence of the first violation, the patron will receive a verbal warning from the RSO. Any other violations or unsafe acts, the perpetrator will be asked to leave the range and the incident(s) will be reported to ODR. At no time will the RSO get into a confrontation with patrons. All situations/incidents should be reported to the 733d Security Forces Squadron (SFS) at (757) 878-4555.

2.3.2. The Fort Eustis Small Arms Range Safety Area (SARSA) Memorandum for Record, dated 28 Feb 17 (Attachment 5), will be executed and adhered to during the conduct of POF Range operations with the following exception: One (1) government employee (RSO) will be the aerial observer, who will be positioned in the range tower. All members of the firing group will be designated as spotters. The RSO is not authorized to fire their POF while on duty. For weekend weather (ceiling/visibility), contact the on-duty USAF Weather Forecaster at (757) 503-0216. The RSO will ensure ceiling and visibility requirements are met in accordance with SARSA requirements.

2.4. Range Safety. All patrons will be briefed on Range Policies and Procedures and will assume Range Safety responsibilities. All patrons will be required to acknowledge/sign the Range Safety Brief. Range Safety will ensure:

2.4.1. Down range firearms effects (bullet impacts) are observed and within the prescribed range limits.

2.4.2. Only approved paper targets are set-up down range.

2.4.3. No personnel or wildlife are down range or any aircraft are visible within the prescribed range limits while firing is being conducted.

2.4.4. Periodic check fires are announced to allow personnel to check their targets down range.

2.4.5. All firearms are cleared and safe before allowing personnel to go down range.

2.4.6. No firearms are being handled while personnel are down range.

2.4.7. No smoking on the firing range.

2.4.8. No running or horseplay on or around the firing line.

2.4.9. All personnel will have ballistic eye and hearing protection when on the firing line.
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2.4.10. A cease fire is called when an unsafe act or condition is observed.

2.5. Clearing the Range. All patrons are responsible for cleaning up after themselves on the POF Range. It is the RSO's duty to ensure that patrons police their area prior to leaving the range. All patrons are responsible for removing their brass and/or expended shells and all trash prior to departure from the range. Brass can be placed in the container labeled "Brass" located at the entrance to the tower. All patrons and guests are required to sign out with the RSO when they are done firing. The RSO will ensure that the POF Range is clear and clean before taking down the range flag.

2.6. Guests. Authorized patrons will be allowed to bring two (2) guests at a time to the POF Range. All guests that do not have base access must be cleared by 733 SFS at 2 Washington Blvd., Fort Eustis (Main Gate Inspection Tent). Sponsors are responsible for ensuring their guests have base access and that any authorized firearm(s) they are using are registered through 733 SFS. Authorized patrons cannot bring guests that are eligible to become patrons. A guest is defined as:

2.6.1. A person that does not have base access, i.e., civilian.

2.6.2. A spouse and/or child age 17 and younger of an authorized patron. If the spouse chooses to fire without his/her sponsor they must acknowledge/sign the Range Safety Brief and obtain a Range Permit.

2.7. Parking. Privately Owned Vehicles (POV) are allowed in designated parking areas only. Parking areas will be outlined in the map provided at ODR when a POF Range Safety Card is purchased.

Chapter 3

AUTHORIZED FIREARMS

3.1. Pistols firing soft point, hollow point, and standard ball, up to and including .45 caliber are authorized.

3.2. Muzzle loaders firing ball and sabot rounds up to and including .58 and using a maximum of 150 grains of black powder propellant are authorized.

3.3. Shotguns are slug only – no birdshot.

3.4. Rifle firers are permitted to fire .22 and .223 (5.56) caliber only. Long, rimfire, standard velocity and match ammunition are allowed for .22 caliber. Ball ammunition only for .223 (5.56).

Chapter 4

RESTRICTIONS

4.1. Restrictions. The following restrictions apply to all patrons:

4.1.1. Alcohol and drugs are prohibited at all times. Any person determined to be under the influence of alcohol or drugs will not be permitted range access.

4.1.2. Wildcat (experimental) handgun and rifle calibers and ammunition are not authorized.

4.1.3. Reloaded ammunition is prohibited.

4.1.4. Tracer ammunition is prohibited.

4.1.5. Firearms which fire automatically or in burst mode are unauthorized. This includes semi-automatic firearms with any modification or attachment which enables more than one round to be fired simultaneously, commonly referred to as “bump firing”.

4.1.6. Only paper targets are approved on the POF Range. Cans, bottles and other objects will not be used as targets.

4.1.7. Harassing or firing at wildlife is prohibited.

4.1.8. Birdshot/bore is prohibited.

Clinton A. Ross, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 36-2654, *Combat Arms Program*, 17 April 2020

ASA FE Regulation 350-1, *Training Support,* 14 February 2011

Prescribed Forms

None

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

AF Form 1314, *Firearms Registration*

DD Form 2760, *Qualification to Possess Firearms or Ammunition*

DD Form 2977, *Deliberate Risk Assessment Worksheet*

Abbreviations and Acronyms

AF – Air Force

AFB – Air Force Base

AFI – Air Force Instruction

AFMAN – Air Force Manual

AFRIMS – Air Force Records Information Management System

ASA-E – Army Support Activity – Eustis

BDOC – Base Defense Operations Center - Eustis

FAAF – Felker Army Airfield

FSD – Force Support Division

MSG – Mission Support Group

NoTAM – Notice to Airmen

ODR – Outdoor Recreation

OPR - Office of Primary Responsibility

POV – Privately Owned Vehicle

POF – Privately Owned Firearm

RSO – Range Safety Officer

SFS – Security Forces Squadron

Attachment 2

Safety and Range Rules

Figure A2.1. Safety and Range Rules

SAFETY/RANGE RULES

Safety is everyone's responsibility and vital while on the range. All patrons authorized to use the range will ensure that all rules are followed. All patrons authorized to use the range must read and sign this Safety/Range Rules sheet.

Firearms must be unloaded with the breech or action open when on the range and not firing, or while persons are down range.

Firearms will remain cased at all times when not on the firing line.

Do not handle firearms while people are down range.

No profane or abusive language.

Shoes and shirt must be worn at all times.

Shoot at approved paper targets only.

Shooters needing to go down range will notify the RSO so the RSO can cease firing on the range. Once the RSO acknowledges the range is clear, firers may then go down range.

Do not go down range unless agreed upon by members of the firing line and the RSO has verified that firearms are safe and cleared. This will be evident when all firearms have been cleared and placed on the ground.

Once all shooters have returned to the firing line, the RSO ensures the range is clear and lifts the cease fire by calling "range is clear".

Shoot only from the firing line and respect the rights of others.

Collect personal brass and clean around your area after finishing your shooting session. Brass can be placed in the container labeled "BRASS" located at the entrance to the tower.

Firearms must be pointed down range at all times and/or in a safe direction.

Do not point any firearm at any person whether the firearm is loaded or unloaded.

Know how your firearm operates.

Ensure your firearm and ammunition is compatible.

All minors 17 years or younger must be accompanied by an adult. While the minor is firing or handling a firearm, the parent or guardian must remain within the immediate area and are not permitted to fire or handle another firearm.

Authorized persons must bring their own paper targets. The following targets are not authorized for use on the range: plastic bottles, aluminum cans, metal targets and glass targets.

All patrons that use the range are designated as spotters and responsible for the safe operation of the range and will ensure the following:

- No firing is conducted when visibility requirements for Controlled Firing Area cannot be met.
- Down range weapon effects are observed to ensure projectiles land within the prescribed range limits.
- The proper targets are set up and down range.
- No personnel or wildlife are down range.
- Announce periodic check fires to permit patrons the opportunity to check targets.
- All weapons are properly cleared prior to permitting patrons down range.
- No weapons are being handled while patrons are down range.
- No smoking on the firing line.
- No running or horseplay on or around the firing line.
- Adequate hearing and eye protection are being used at all times.

Any patron observing an unsafe act must call for a cease fire.

The range must be cleaned prior to leaving. Before leaving the range, all shooters must pick up their spent casings and remove targets from the target stands. For safety reasons, there are no trash cans on the range - you must take your trash with you.

733 FSD Representative (Print)	Authorized Patron (Print)
733 FSD Representative (Signature)	Authorized Patron (Signature)

Attachment 3

Release and Hold Harmless Agreement.

Figure A3.1. Release and Hold Harmless Agreement.

RELEASE AND HOLD HARMLESS AGREEMENT

I, _____, for and in consideration for being permitted to use the POF Range on Fort Eustis, hereby for myself, my heirs, executors or administrators, waive and release any and all rights and claims for damage or liability I may have against the United States of America, its agencies, Soldiers and employees, for any and all personal injuries or property damage I may suffer while participating in said activities. I certify that I am medically fit to participate in these activities. I agree that there are numerous risks involved in the undertaking of these activities. By participating in these activities, I hereby agree to assume the risks of any bodily injury whatsoever and declare that I enter into the agreement freely and voluntarily. I further agree that I or my estate shall indemnify the United States Government for any and all cost of defending any suit or administrative action brought by me, my heirs or estate in violation of this agreement.

I understand and agree to the following:

I have received a safety briefing concerning the use of the POF Range.

I am responsible for obeying all rules and regulations of the POF Range.

I will fire at no targets other than those approved. I will not fire at rocks, bottles, signs, cans or other unauthorized targets.

I understand this wavier will expire one (1) year from date issued.

_____	_____
733 FSD Representative (Print)	Authorized Patron (Print)
_____	_____
733 FSD Representative (Signature)	Authorized Patron (Signature)
_____	_____
Date of Agreement	Date of Agreement
_____	_____
Date Agreement Expires (One (1) Year)	Date Agreement Expires (One (1) Year)

Attachment 4

Release and Hold Harmless Agreement for Minor.

Figure A4.1. Release and Hold Harmless Agreement for Minor.

RELEASE AND HOLD HARMLESS AGREEMENT (MINOR)

I, _____ and _____, parent(s) of a minor child, and _____, a minor, in consideration for being permitted to utilize the POF Range, hereby for myself (ourselves), my (our) heirs, executors or administrators, waive and release any and all rights and claims for damage or liability that I (we) may have against the United States of America, its agencies, Soldiers and employees, for any and all personal injuries or property damage I (we) or my (our) minor child may suffer while participating in said activities. I (we) certify that my (our) minor child is medically fit to participate in these activities. I (we) certify that my (our) minor is under the age of 18.

I (we) agree that there are numerous risks involved in undertaking their activities. By participating in these activities, I (we) hereby agree to assume the risks of any bodily injury whatsoever and declare that I (we) enter into this agreement freely and voluntarily.

I (we) further agree that I (we) or my (our) estate shall indemnify the United States Government for any and all costs of defending any suit or administrative action brought by me (us) or my (our) heirs or estate in violation of this agreement.

I (we) understand and agree to the following:

I (we) have received a safety briefing concerning the POF Range and the range operations.

I (we) am responsible for obeying all rules and regulations of the POF Range.

I (we) will fire at no targets other than those approved.

I (we) understand this waiver will expire one (1) year from date of issue.

Minor Child Name (Print)	Date of Birth
733 FSD Representative (Print)	Authorized Patron(s) (Print)
733 FSD Representative (Signature)	Authorized Patron(s) (Signature)
Date of Agreement	Date of Agreement

Attachment 5

Small Arms Range Safety Area (SARSA) Memo.

Figure A5.1. Small Arms Range Safety Area (SARSA) Memo.

DEPARTMENT OF THE ARMY
US ARMY SUPPORT ACTIVITY, FORT EUSTIS
705 WASHINGTON BLVD
FORT EUSTIS, VIRGINIA 23604-1278

IMEV-PLT 28 February 2017

MEMORANDUM FOR RECORD

SUBJECT: Small Arms Range Safety Area (SARSA) for FEVA Range Complex 9mm, .45 Cal, .38 Cal Ball, 12 Gauge Slug, 5.56mm (M193), Ball, 5.56mm (196) Tracer, 5.56mm (M855) Ball, 5.56mm (856) Tracer, 5.56mm(862) Plastic, 7.62mm (118), .50 Cal M2 AP, .50 Cal (M858) Tracer Ball, .50 Cal (M860) Tracer Ball Plastic, .50 Cal M2 Ball, 20 mm M220, 30MM (M789) HEDP.

1. References:

a. DA Pam 385-63, Range Safety, 16 April 2014.

b. AR 385-63, Range Safety, 30 January 2012.

c. AR 95-2, Airspace, Airfield/Heliports, Flight Activity, Air Traffic Control, and Navigational Aids, 10 April 2007

2. A SARSA is established for the Fort Eustis Range Complex to protect non-participating aircraft from activities conducted on the Range Complex. The Fort Eustis Range Complex does not fall within a Restricted Area or Controlled Firing Area. This SARSA was developed in accordance with the requirements outlined in DA PAM 385-63, Range Safety, 16 April 2014.

3. SARSA operating hours will be from Monday through Friday, 0700-1600, or until all live fire ranges have completed firing, and on Saturday and Sunday, upon units request, from 0700-1600 or until all live fire ranges have completed firing.

4. The primary means of communication between Felker Army Airfield and range control will be hand held radio to base station channel 1, and secondary means will be land-line at (757)878-4412 ext 226/234, Monday through Friday, 0700-1600 or until all live fire ranges have completed firing. During weekend training, all military aircraft approaching or needing entrance into the SARSA area will contact Range Operations for clearance on FM frequency 32.00.

5. The Fort Eustis Range Complex is located on Mulberry Island Road at eastern boundary UG57350900-southeastern boundary UG61780595 and western boundary UG60620843- southwestern boundary UG 60650495. Firers will stay within the posted

left and right limits established on the range (see attached map of the Surface Danger Zone (SDZ), (Encl 1).

6. The Range Safety Officer (RSO) will brief all personnel conducting training within the Range Complex, of potential dangers of aircraft entering the SARSA area, and the required cease fire procedures. The using unit will designate two aerial observers for the range being utilized, who will be posted in the range tower to monitor all air traffic. Any aircraft lower than the SARSA vertical hazard and inside a weapon's respective surface danger zone is considered in danger of the vertical hazard and will necessitate an immediate cease fire.

7. The Officer in Charge (OIC) of the range is responsible for monitoring the skies for aircraft out to 5 miles from the boundary of the SDZ of the SARSA. The OIC will call an immediate cease fire should an aircraft appear to be approaching or penetrating the SDZ. Ranges cannot be used when the visibility or cloud height becomes less than the vertical hazard of the weapon being fired.

8. Any changes to the type of activity or modification of the range will require new SARSA establishment approved by the commander. Changes will be coordinated with the Department of the Army Representative to the FAA (DAR) prior to signatures. Any deviation will be processed in accordance with AR 385-63, Paragraph 3-1.

9. The active deviation for the SDZ for the range utilized by Aviation Applied Technology Directorate will remain in effect as approved, (Encl 2).

10. For further information contact the Range Safety Officer, Willie Scott, at (757) 878-7342.

// digitally signed 7 March 2017 //

JAMES W. MCKENNA
LTC, LG
Commanding

Encls
SDZ of Ranges
Deviation for Aviation Applied Technology Directorate

Figure A5.2. Surface Danger Zone (SDZ) Memo.

DEPARTMENT OF THE ARMY
HEADQUARTERS, UNITED STATES ARMY CENETR FOR INITIAL MILITARY TRAINING
201 DILLON CIRCLE
FORT EUSTIS, VIRGINIA 23604-5701

REPLY TO ATTENTION OF:

FEB 07 2017

ATMT

MEMORANDUM FOR Director, Aviation Applied Technology Directorate (RDMR-ADA/COL Braddom), US Army Research, Development and Engineering Command, Building 401, Lee Boulevard, Ft. Eustis, VA 23604-5577

SUBJECT: Deviation of Surface Danger Zone (SDZ) Dimensions for Firing of Small Caliber up to and Including 30mm High Explosive (HE) and High Explosive Incendiary (HEI) Ammunition

1. Your request for SDZ deviation required by AR 385-63 is approved subject to the following firing restrictions:

a. All weapon firing will be done at zero, ±5º elevation so that the trajectory is never above the 30-foot horseshoe shaped berm.

b. All weapon firing will be from fixed mounts with the exception of 7.62mm and 5.56mm weapon firing which on occasion will be shoulder fired.

c. A steel plate will be positioned behind the target for all HEI firings to assure projectile detonation should it penetrate through the target.

d. All firing will be coordinated through Range Control.

e. All firing will be conducted IAW the Ballistic Test Range for Aircraft Component Survivability (BTRACS) Standing Operating Procedures under the guidance of Aviation Applied Technology Directorate Supervisors.

f. All other test range relevant safety requirements contained in DA Pam 385-63, ASA Regulation 350-1, and the risk assessment will be met.

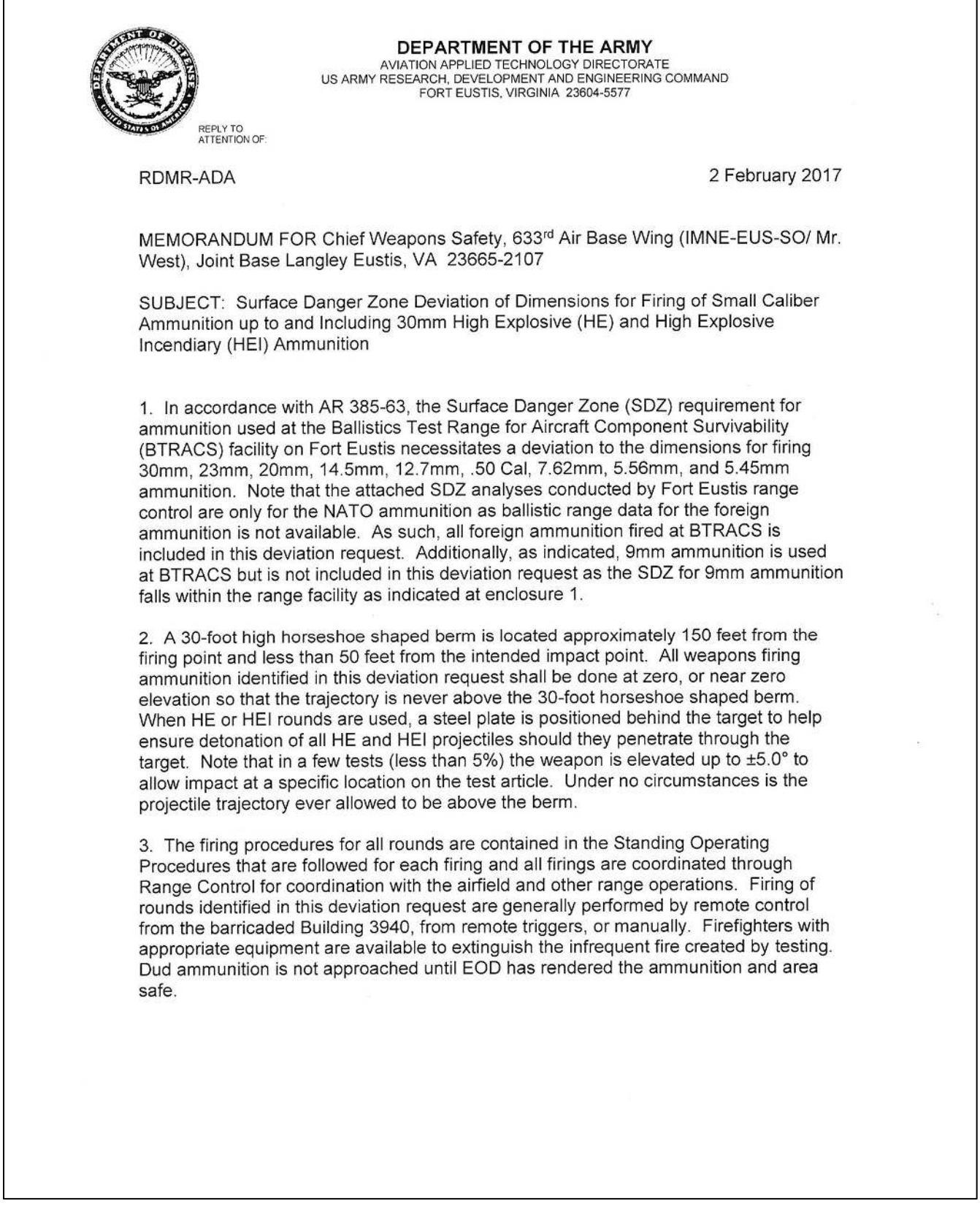
2. This deviation is valid for a period of 1-year from the date of this memorandum. In the event of an accident or incident related to this SDZ deviation during use of this range under these controlled conditions, this deviation becomes invalid until reinstated by this headquarters.

3. The point of contact for this action is Mr. Duane West, Chief Weapons Safety at (757) 764-5059.

// digitally signed 2 February 2017 //

ANTHONY C. FUNKHOUSER
Major General, U.S. Army
Senior Commander - Army Element

Figure A5.3. Chief Weapons Safety Memo.



DEPARTMENT OF THE ARMY
AVIATION APPLIED TECHNOLOGY DIRECTORATE
US ARMY RESEARCH, DEVELOPMENT AND ENGINEERING COMMAND
FORT EUSTIS, VIRGINIA 23604-5577

REPLY TO
ATTENTION OF:

RDMR-ADA

2 February 2017

MEMORANDUM FOR Chief Weapons Safety, 633rd Air Base Wing (IMNE-EUS-SO/ Mr. West), Joint Base Langley Eustis, VA 23665-2107

SUBJECT: Surface Danger Zone Deviation of Dimensions for Firing of Small Caliber Ammunition up to and Including 30mm High Explosive (HE) and High Explosive Incendiary (HEI) Ammunition

1. In accordance with AR 385-63, the Surface Danger Zone (SDZ) requirement for ammunition used at the Ballistics Test Range for Aircraft Component Survivability (BTRACS) facility on Fort Eustis necessitates a deviation to the dimensions for firing 30mm, 23mm, 20mm, 14.5mm, 12.7mm, .50 Cal, 7.62mm, 5.56mm, and 5.45mm ammunition. Note that the attached SDZ analyses conducted by Fort Eustis range control are only for the NATO ammunition as ballistic range data for the foreign ammunition is not available. As such, all foreign ammunition fired at BTRACS is included in this deviation request. Additionally, as indicated, 9mm ammunition is used at BTRACS but is not included in this deviation request as the SDZ for 9mm ammunition falls within the range facility as indicated at enclosure 1.

2. A 30-foot high horseshoe shaped berm is located approximately 150 feet from the firing point and less than 50 feet from the intended impact point. All weapons firing ammunition identified in this deviation request shall be done at zero, or near zero elevation so that the trajectory is never above the 30-foot horseshoe shaped berm. When HE or HEI rounds are used, a steel plate is positioned behind the target to help ensure detonation of all HE and HEI projectiles should they penetrate through the target. Note that in a few tests (less than 5%) the weapon is elevated up to ±5.0° to allow impact at a specific location on the test article. Under no circumstances is the projectile trajectory ever allowed to be above the berm.

3. The firing procedures for all rounds are contained in the Standing Operating Procedures that are followed for each firing and all firings are coordinated through Range Control for coordination with the airfield and other range operations. Firing of rounds identified in this deviation request are generally performed by remote control from the barricaded Building 3940, from remote triggers, or manually. Firefighters with appropriate equipment are available to extinguish the infrequent fire created by testing. Dud ammunition is not approached until EOD has rendered the ammunition and area safe.

RDMR-ADA
SUBJECT: Surface Danger Zone Deviation of Dimensions for Firing of Small Caliber Ammunition up to and Including 30mm High Explosive (HE) and High Explosive Incendiary (HEI) Ammunition

4. Request a 1-year deviation of the SDZ dimensions for operations at the BTRACS facility, provided at enclosure 1. The original overlay of this SDZ was generated by and is being retained and maintained by the Range Control Office for posting on the installation range maps. Cone surface danger zones are used for the rounds identified in this deviation request due to all firings being controlled within the 30-foot horseshoe shaped berm.

5. The Deliberate Risk Assessment Worksheet developed for this action by Mr. Donald Campbell, BTRACS Range Officer, on 26 November 2016 is provided with this deviation request at enclosure 2.

6. My point of contact for this action is Mr. Peter Montrond, RDMR-ADA-D, Directorate Safety Officer, 878-5888 or Mr. Donald Campbell, 878-2868, BTRACS Range Officer.

// digitally signed 2 February 2017 //

2 Encls

STEVEN R. BRADDOM
COL, AV
Director

Figure A5.4. Example of DD Form 2977, Deliberate Risk Assessment Worksheet.

DELIBERATE RISK ASSESSMENT WORKSHEET

1. MISSION/TASK DESCRIPTION	2. DATE (DD/MM/YYYY)
BTRACS Weapons Testing	03/02/2017

3. PREPARED BY

a. Name (Last, First Middle Initial)	b. Rank/Grade	c. Duty Title/Position
Campbell, Donald, A.	DE-03	Range Officer

d. Unit	e. Work Email	f. Telephone (DSN/Commercial (Include Area Code))
AATD	donald.a.campbell.civ@mail.mil	757-878-2868

g. UIC/CIN (as required)	h. Training Support/Lesson Plan or OPORD (as required)	i. Signature of Preparer
W1DF13	As per applicable Test Plan	// digitally signed 3 February 2017 //

Five steps of Risk Management: (1) Identify the hazards (2) Assess the hazards (3) Develop controls & make decisions (4) Implement controls (5) Supervise and evaluate *(Step numbers not equal to numbered items on form)*

	4. SUBTASK/SUBSTEP OF MISSION/TASK	5. HAZARD	6. INITIAL RISK LEVEL	7. CONTROL	8. HOW TO IMPLEMENT/ WHO WILL IMPLEMENT	9. RESIDUAL RISK LEVEL
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	Rounds landing outside the SDZ	H	Firing from set fixture target. 30' horseshoe shape berm approximately 150' from the firing point. Firing fixture never elevated (<5 degrees) to allow a trajectory above the berm.	How: Follow SOP and operate from an approved range deviation. Who: R&D Technician Direct supervisor Test personnel	M
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	Excessive Noise	M	All personnel will wear appropriate hearing protection	How: Follow SOP and operate from an approved range deviation. Who: R&D Technician Direct supervisor Test personnel	L
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	Personnel down range	H	All firing is coordinated through range control. All range personnel wear suitable hearing protection	How: Follow SOP and operate from an approved range deviation. Who: R&D Technician Direct supervisor Test personnel	M

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	4. SUBTASK/SUBSTEP OF MISSION/TASK	5. HAZARD	6. INITIAL RISK LEVEL	7. CONTROL	8. HOW TO IMPLEMENT/ WHO WILL IMPLEMENT	9. RESIDUAL RISK LEVEL
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	Aircraft overflight	M	Adhere to FLIPS. Coordinate all firing through range control. Follow cease fire procedures. Follow SARSA	How: Airfield will be notified when R&D range is in a hot status by range control Who: R&D Technician Direct supervisor Test personnel	L
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	Improper weapons handling	H	Only well trained experienced operators will conduct operations.	How: Follow SOP and operate from an approved range deviation. Who: R&D Technician Direct supervisor Test personnel	M
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	Ammunition Duds	H	Keep personnel away from duds. Call EOD for removal and notify range control of the incident	How: Follow SOP and operate from an approved range deviation. Who: R&D Technician Direct supervisor Test personnel	M
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	Fires at target location	M	Fire fighters with appropriate equipment on standby to fight fires	How: Follow SOP and operate from an approved range deviation. Who: R&D Technician Direct supervisor Test personnel	L
+ -	Firing all calibers from 5.45 through 30MM to included all NATO and foreign ammunition	HEI ammunition not detonating	H	Steel plate is placed in the shot line, behind the target to ensure that the projectile detonates or is destroyed upon impact	How: Follow SOP and operate from an approved range deviation. Who: R&D Technician Direct supervisor Test personnel	M

10. OVERALL RESIDUAL RISK LEVEL *(All controls implemented)*:

☐ EXTREMELY HIGH ☐ HIGH ☒ MEDIUM ☐ LOW

11. OVERALL SUPERVISION PLAN AND RECOMMENDED COURSE OF ACTION

Direct supervisor will ensure that all test personnel and technicians are briefed on controls prior to test events. Range OIC and Safety Officer will ensure that all personnel on the range strictly adhere to range safety procedures IAW applicable Army and Air Force regulations.

12. APPROVAL OR DISAPPROVAL OF MISSION OR TASK ☒ Approve ☐ Disapprove

a. Name (Last, First, Middle Initial)	b. Rank/Grade	c. Duty Title/Position	d. Signature of Approval Authority
Montrond, Peter B.	DE-04	Directorate Safety Officer	// digitally signed 3 February 2017 //

e. Additional Guidance:

Risk Assessment Matrix		Probability *(expected frequency)*				
		Frequent: Continuous, regular, or inevitable occurrences	**Likely:** Several or numerous occurrences	**Occasional:** Sporadic or intermittent occurrences	**Seldom:** Infrequent occurrences	**Unlikely:** Possible occurrences but improbable
Severity *(expected consequence)*		A	B	C	D	E
Catastrophic: *Mission failure, unit readiness eliminated; death, unacceptable loss or damage*	I	EH	EH	H	H	M
Critical: *Significantly degraded unit readiness or mission capability; severe injury, illness, loss or damage*	II	EH	H	H	M	L
Moderate: *Somewhat degraded unit readiness or mission capability; minor injury, illness, loss, or damage*	III	H	M	M	L	L
Negligible: *Little or no impact to unit readiness or mission capability; minimal injury, loss, or damage*	IV	M	L	L	L	L

Legend: **EH** - Extremely High Risk **H** - High Risk **M** - Medium Risk **L** - Low Risk

13. RISK ASSESSMENT REVIEW *(Required when assessment applies to ongoing operations or activities)*				
a. Date	b. Last Name	c. Rank/Grade	d. Duty Title/Position	e. Signature of Reviewer
14. FEEDBACK AND LESSONS LEARNED				
15. ADDITIONAL COMMENTS OR REMARKS				

Figure A5.5. Surface Danger Zone (SDZ) Analysis Chart.

